

ESDS AI Innovation Call

Call for Ideas — FY2026

Issued by: Earth Science Data Systems Issue Date: **May 5, 2026** Innovation Calls Due: **Jun 5, 2026**

1. Purpose and Scope

1.1 Purpose

NASA's Earth Science Data System (ESDS) program seeks ideas for projects that advance the [ESDS AI Strategy](#) by developing AI-powered solutions or enhancements to meet challenges in the Earth Science Data System. Ideas must clearly define, align with the strategy, and justify the specific challenge in the data system and how the project will address that challenge. Priority will be given to ideas that demonstrate enhancements to existing applications or systems and have clear paths to operationalization. We highly encourage ideas that demonstrate how AI can make existing projects faster, smarter, or more useful.

This is an open Call within ESDS: we do not seek projects addressing any specific use cases. The best ideas will come from the people closest to the work. As such, project teams must be led by or include members from the core ESDS portfolio (e.g., ESDIS, ODSI, SIPS, DAACs).

1.2 Scope

NASA's Earth Science Data Systems (ESDS) program exists to make NASA's free and open Earth science data interactive, interoperable, and accessible for research and societal benefit. ESDS manages nearly 180 petabytes of Earth science data today, a number expected to exceed 600 petabytes by the early 2030s.

That scale of growth outpaces what human attention alone can manage. In FY26, ESDS released its Artificial Intelligence (AI) strategy which defines how the program applies AI across four pillars of the NASA Earthdata lifecycle: Infrastructure, Production, Access and Analysis. The strategy emphasizes two underlying cross-cutting capabilities: Governance and Standards, and Operational Intelligence. This Call is one of the five tactics in that strategy. It will fund practical, near-term AI work across the ESDS ecosystem — with a strong preference for enhancements to existing projects over greenfield development.

2. Eligibility

The innovation call is open to any funded, directed effort within the Earth Science Division where there is a clear connection to NASA Earth science data access, production, or use. Eligible existing efforts include, but are not limited to:

- ESDS-funded data systems teams and projects (e.g., ESDIS, ODSI, SIPS, DAACs)
- Mission-directed data system efforts with active ESDS engagement
- Directed projects within ESD with a data access, production, or stewardship component

While ideas from teams outside the core ESDS portfolio are welcome, they must clearly demonstrate a strong connection to ESDS, the AI strategy pillars, and **must** include at least one team member from the core ESDS portfolio. When in doubt, reach out before submitting.

Ideas must be submitted by a funded team lead with appropriate institutional authorization. Collaborations across teams are strongly encouraged.

3. Requirements

All ideas must demonstrate alignment with the ESDS AI strategy and connect clearly to at least one of the four pillars in the ESDS AI Strategy (Infrastructure, Production, Access, Analysis). Ideas must also include the following:

- A clear articulation of the problem being solved and why AI is the right approach
- Description of how the proposed work connects to operational use — what system, workflow, or community will benefit and how
- A plan for how the prototype or enhancement will be validated, and by whom
- A statement of how outputs will be made openly available consistent with NASA open science policy
- Identification of any dependencies on external tools, models, or platforms, and any associated data rights or licensing considerations

3.1 Budget and Duration

Funds under this call are expected to be in the range of \$150,000–\$200,000 per project. Approximately A limited number of funds (5-10 ideas will be funded) will be made subject to available resources (\$1M total). Funded project periods must begin no later than September 2026 and must not exceed 12 months in duration.

Budget narratives (see Appendix A.) should include direct costs of AI development work, including personnel, compute, and tooling. Indirect costs are allowable per existing funding terms. Budgets must not include requests for hardware procurement or travel unless directly necessary and justified.

3.2 Innovation Call Format and Page Limit

Calls must follow the template provided in Appendix A of this document. Submissions not meeting these requirements will not be reviewed.

Page limit: 6 pages (excluding cover page, budget table, work effort table and references)

Font: Arial 11pt, 1-inch margins

File format: PDF

Submission method: [Google Form](#)

4. Summary of Key Information

Total amount of funding Available	\$1,000,000
Expected budget range for funded ideas	\$150,000 to \$200,000
Anticipated number of funds	5-10
Due date for idea submission	Jun 5, 2026
Notification of status sent to proposers	NLT Jul 10, 2026
Project period begins	No later than Sep 1, 2026
Project duration	No more than 12 months

4.1 Deliverables

All funded projects are expected to deliver:

- A working prototype or integration demonstrating the proposed AI capability
- A brief final report (<10 pages) covering: what was built, how it was validated, lessons learned, and clear path(s) to operational status
- An open-source code repository or equivalent artifact, documented and accessible
- A presentation including project overview and a technology demonstration to the Earth Data Officer and their leadership team

4.2 Evaluation Criteria

Innovation calls will be evaluated on a 100-point scale across five criteria.

Criterion	Description	Points
Strategic alignment	The idea connects clearly to the ESDS AI Strategy and identifies at least one pillar (Infrastructure, Production, Access, or Analysis) the work advances. The connection is specific and substantive.	25
Problem definition and AI justification	The problem is clearly defined and the proposed use of AI is well-motivated. The idea demonstrates an understanding of what type of AI capability is being proposed and is candid about its limitations and risks.	20

Operational relevance	The idea identifies a specific system, workflow, or user community that will benefit, and describes a plausible path from prototype to operational use within ESDS systems.	25
Validation and open science	The idea includes a credible validation plan with defined success criteria and an identified validator. Any dependencies on external tools, models, or data with licensing implications are disclosed and addressed.	15
Team capability and budget	The team demonstrates the capacity to execute within the project period. The budget is realistic and clearly tied to the proposed scope, with personnel, compute, and tooling costs appropriately justified.	15
Total		100

Appendix A: Innovation Call Template

All ideas must follow this template. Do not add sections or reorganize. Evaluators will use this structure to assess submissions consistently.

Innovation Call Title:

Submitting Team / Organization:

Team Lead Name and Contact:

ESDS Pillar(s) Addressed: Infrastructure / Production / Access / Analysis

Requested Funding:

A.1 Overview and Background

Describe the problem you are solving. What is the current state? Why does it matter for Earth science data systems? What has already been tried or considered? (Suggested: 1 pages)

A.2 Proposed Approach

Describe your AI-powered solution. What will you build or enhance? What AI tools, models, or frameworks will you use and why? How does this connect to an existing project or operational system? (Suggested: 1–1.5 pages)

A.3 Limitations and Challenges

Be honest about what could go wrong, what is uncertain, and what dependencies exist. How will you manage or mitigate these? (Suggested: 0.5 pages)

A.4 Deliverables

List the specific outputs you will produce: working prototype, integration, documentation, code repository. Be concrete. What will exist at the end of the project that does not exist today? (Suggested: 0.5 pages)

A.5 Validation and Path to Operations

How will you know it works? Who will test or validate the output? What would need to happen for this to move from prototype to operational use? (Suggested: 0.5 pages)

A.6 Timeline

Provide a month-by-month, milestone-based schedule for the project duration. Include key decision points and the final deliverable date.

A.7 Team

List the people who will do the work, their roles, and the estimated effort in person-months. Include any key collaborators or external dependencies.

A.8 Budget Summary

Provide a high-level budget breakdown: personnel, compute, other direct costs. One page or less.

Appendix B:. Frequently Asked Questions (FAQ)

1. What is the primary goal of the ESDS AI Innovation Call?

The call is designed to fund practical, near-term AI work across the ESDS ecosystem. It invites funded, directed efforts within the Earth Science Division to propose AI-powered solutions to real problems in Earth science data systems.

2. Is there a preference for new projects or existing enhancements?

Enhancements to existing projects are strongly encouraged and are the preferred starting point over greenfield development. The goal is to apply AI to make existing work faster, smarter, or more useful.

3. Who is eligible to submit an idea?

Innovation calls are open to any funded, directed effort within the Earth Science Division that has a clear connection to Earth science data access, production, or use.

- ESDS-funded data systems teams and projects (DAACs, SIPs, ESDIS projects)
- Mission-directed data system efforts with active ESDS engagement
- Directed projects within ESD with a data access, production, or stewardship component

Innovation calls must be submitted by a funded team lead.

4. What is the expected funding range per project?

Funds are expected to be in the range of **\$150,000–\$200,000** per project. It is expected there will be 5-10 total projects.

5. What costs are covered by the funding?

Funding covers the direct costs of AI development work, including personnel, compute, and tooling. Indirect costs are also allowable per existing funds terms.

6. What costs are NOT covered?

Funding does not cover hardware procurement, new infrastructure standing, or travel, unless the travel is directly necessary and justified.

7. What are the key programmatic requirements for the idea?

All ideas must demonstrate alignment with the ESDS AI Strategy and connect clearly to at least one of the four pillars (Infrastructure, Production, Access, Analysis).

- A clear articulation of the problem being solved and why AI is the right tool
- A description of how the proposed work connects to operational use (e.g., what system or community benefits)
- A plan for how the prototype will be validated, and by whom
- A statement of how outputs will be made openly available consistent with NASA open science policy
- Identification of any external dependencies (tools, models, platforms), and associated data rights or licensing considerations

8. What are the mandatory deliverables for selected projects?

All selected projects are expected to deliver:

- A working prototype or integration demonstrating the proposed AI capability
- A brief final report (<10 pages) covering what was built, validation, lessons learned, and recommended path to operational status
- An open-source code repository or equivalent artifact, which must be documented and accessible
- A presentation and tech demonstration to the Earth Data Officer and their leadership team